

Lightweight object detection in low light: Pixel-wise depth refinement and TensorRT optimization

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ABSTRACT

Images captured in low-light conditions present significant challenges for accurate object detection due to factors such as high noise, poor illumination, and low contrast. In this study, we propose an innovative approach that combines pixel-wise depth refinement and TensorRT optimization to improve object detection performance and inference speed under challenging lighting scenarios. The Pixel-wise depth refinement utilizes a deep learning model to estimate the depth curve for each pixel in the low-light image, providing detailed scene depth information. We used the Lightweight YOLOv8 object detection model using TensorRT. This will reduce the model's computational complexity and memory footprint, enabling faster inference and real-time performance on resource-constrained devices. By combining these techniques, our approach aims to accomplish enhanced entity recognition precision in reduced-illumination environments while maintaining efficiency and real-time performance. We evaluate the effectiveness of our proposed modifications to the depth curve estimation block and the overall approach on benchmark datasets and demonstrate its potential for enhancing object detection capabilities in challenging low-light conditions. Furthermore, the experimental results show an improvement in the precision, recall, and mean average precisions (mAP), and also by using the TensorRT the inference speed is four times faster than the existing one.

1. Introduction

As technology becomes more advanced, autonomous vehicles are becoming incredibly popular and accessible for consumers across countries and economic levels. Increasing accessibility leads to safer transportation, improved traffic flow, and increased economic and productivity. To ensure driver safety, highly efficient and accurate learning models must be trained on various driving scenarios to accurately recognize objects in various weather and lighting conditions accurately [1]. Autonomous Vehicle Perception (AVP) refers to a system's ability to extract vital information from its environment. Autonomous driving relies on accurate information about the driving environment, such as free drivable regions, locations, velocities, and predicted impediments. Autonomous vehicles employ LiDAR and camera sensors to detect obstructions [2] and avoid accidents. Deep learning algorithms have become the leading approach in remote sensing picture feature recognition because of their efficient image separation abilities. These approaches may be categorized as regression-oriented and region proposal-oriented identification approaches. Regression-oriented

techniques like SSD and the YOLO series [3–6] provide rapid prediction times and are suitable for instantaneous identification applications. The other category includes R-CNN [7–11], and its variants which rely on identifying potential areas of interest. When detecting objects in an image with extremely low light conditions Deep Learning object detection algorithms [12] face certain issues like deformation, viewpoint variation, illumination conditions, occlusion, etc. Low light circumstances may occur based on the time of day and the availability of light sources. The influence of these image changes on state-of-the-art algorithms can be assessed by comparing their performance on similar datasets and images.

The human eye's extraordinary adaptation allows us to easily navigate even poorly illuminated settings. However, for machines entrusted with object detection, the lack of light poses a significant barrier. The shadows become a hiding spot, absorbing the curves and nuances that aid in correct identification. This research will get into the complexities of object detection in low-light environments, examining limitations, current techniques, and promising pathways for obtaining strong performance in the absence of illumination. Traditional object detection

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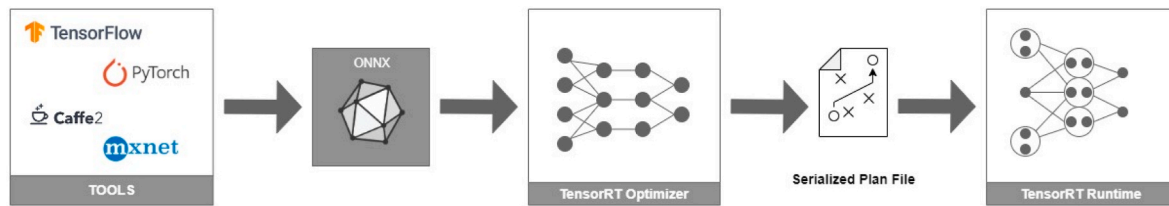


Fig. 1. The workflow of open neural network exchange (ONNX).

algorithms that were trained on high-quality data tend to fail in low-light conditions. Techniques such as histogram equalization [13] and image brightness [14] can provide minor gains, but they also risk magnifying noise and distorting image connections. More complex techniques, such as pre-trained models with adaptive thresholding or region-based enhancements, show promise, but they are computationally expensive and lack generalizability across a wide range of low-light circumstances.

1.1. Low-light enhancement for object detection

Low-light enhancement refers to an image processing technique [15] designed to improve the visibility and clarity of captured images in low-light circumstances. These approaches address the issues of low signal-to-noise ratio, weakened object features, and inappropriate color information that occurs when light is sparse. The goal is to improve characteristics while avoiding undesired artifacts or distortions. Low light enhancement during tough weather conditions adds an element of complication. Rain, snow, and fog further reduce vision, bringing additional issues such as scattered light, color distortions, noise aberrations, etc. Object recognition is a computer vision image problem that consists of recognizing and locating things in images or videos. It is an essential component of numerous applications, including autonomous vehicles, robotics, and security systems. However, object detection might be difficult in inclement weather circumstances such as rain, snow, fog, and poor lighting. These conditions can decrease the quality of videos and images, making it harder for object detectors to precisely identify and locate objects [16,17]. Here's where low-light enhancement utilizes techniques like traditional image processing or deep learning to improve the quality of low-light images. Enhanced images provide clearer details and reduced noise, making it easier for object detection models to identify and classify objects accurately.

1.2. Overview of ONNX and TensorRT

The Open Neural Network Exchange (ONNX) [18] is a freely available AI environment. Various technical companies and academic institutions have developed an open standard for articulating machine learning algorithms and programming solutions to promote research and cooperation in the field of artificial intelligence. ONNX strives for compatibility. It establishes a standard collection of operators and a universal data structure for ML and DL models, enabling AI professionals to employ patterns throughout various applications, devices, developers, and duration. The workflow of ONNX is shown in Fig. 1, it supports several software frameworks, including PyTorch, TensorFlow, Apache MXNet, and Caffe2. ONNX also allows for model optimization on a variety of system appliances. Clients may utilize an ONNX framework with customized intervals for individual systems to enhance the speed of deduction implementation on the unit. TensorRT [19] is a powerful inference optimizer and runtime for deep learning models. It may use ONNX models and optimize them for certain hardware and precision settings, like GPUs and INT8. It can also include capabilities like dynamic shapes, plugins, and quantization-aware training. This paper is organized as follows. Section 2 reviews related work and the current literature in the field. Section 3 details the proposed method and

solution. Section 4 presents the simulation results and relevant comparisons. Finally, Section 5 concludes the paper.

2. Related works

2.1. Low-light enhancement approaches

The [20] investigates the effectiveness of various deep learning models for enhancing low-light natural scene images. It takes a two-pronged approach. First, it compares the quality of the enhanced images produced by different deep-learning models. And further by analyzing how these enhancements affect the accuracy of object detection and classification tasks. Deep learning methods, such as Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs), are often used to improve low-light Images [21]. Deep learning-based low-light image enhancement techniques are methodically categorized and summarized in this work, which also offers resolves for current methodological issues and thorough insights into the fundamental concepts of these algorithms. To ensure high stability and reproducibility, image enhancement techniques are evaluated quantitatively using metrics such as SSIM (Structural Similarity Index Measure) and PSNR (Peak Signal-to-Noise Ratio). In Ref. [22] they explore the world of deep learning algorithms for low-light image enhancement. It delves into various approaches used to improve the quality of images captured in poor lighting conditions. The survey categorizes these deep learning algorithms into different groups as Supervised Learning Methods, Unsupervised Learning Methods, Semi-Supervised Learning Methods, and Zero-Shot Learning Methods. They likely explore the specific functionalities and advantages of each category, providing a comprehensive overview of the deep learning landscape for low-light image enhancement. The most prevalent low-light image-enhancing technique is Gamma transformation [23] This technique modifies an image's brightness and contrast by applying a power-law function to each pixel value. Histogram equalization increases an image's contrast by shifting pixel values to create a consistent histogram [24,25]. An image enhancement method for identifying prominent objects in low light photos is proposed in Refs. [26,27]. The physical lighting model is integrated into a deep neural network by the model, which interprets ambient light as a point-wise variate that varies based on local content. A Non-Local-Block Layer records the variation between neighborhood-favoring regions and local content. Promising outcomes have been observed on four public datasets and one benchmark dataset using this strategy. The Enlight-enGAN [28] is an unconstrained creative stochastic network that can be developed without low/normal-light image combinations and performs very well on a range of real-world test photographs. Retinex methods [29] are techniques derived from the Retinex theory, which proposes that an image may be segmented into reflectance and light elements. The goal is to improve the reflectance while retaining the lighting component. Machine learning and deep learning techniques [30] These methods use data-driven methodologies to determine the relationship between low-light and improved photos. They can use multiple designs, such as autoencoders, generative adversarial networks, or transformers, to get better outcomes than standard methods. In Ref. [31] for unstable lighting conditions they used a homomorphic filtering. The goal of the homomorphic filtering



Fig. 2. Sample Low-light images from the datasets.

framework is to achieve linear separability for illumination-induced noise. The [32] suggests LLHFNet (Low-Light Homomorphic Filtering Network), a novel LLI enhancement model that integrates easily into classification models and does image-to-frequency filter learning. The model maximizes the performance of both image enhancement and classification.

2.2. Deep learning object detection methods

Autonomous vehicles require precise computer vision systems to navigate and localize themselves in severe climatic circumstances. A study combined the YOLOv7 and ESRGAN deep learning approaches to improve detection performance. ESRGAN compensates for poor weather conditions in images, improving detection performance. The hybrid approach achieved 80 % accuracy in both good and bad weather [33–35]. In Refs. [36,37] examines cutting-edge object detection technologies for autonomous vehicles (AVs), focusing on sensory systems and algorithms. It describes autonomous driving operations, various sensory systems used in AVs, and the role of Deep Neural Networks (DNN) in object detection tasks. The evaluation found future research into vehicles with higher levels of automation, emphasizing the importance of object detection in autonomous driving tasks [38]. details a deep learning sensing system for autonomous vehicles that categorizes climate circumstances as either detrimental or acceptable. The framework evaluates three effective deep convolutional neural networks (CNNs) ResNet-50, SqueezeNet, and EfficientNet, through transfer learning approaches. The models were tested on two weather imaging

datasets, DAWN2020 and MCWRD2018, and all of them outperformed the others. Detection-driven Enhancement Network (DENet) [39] uses a Laplacian pyramid to break down images into low and high-frequency components. Separate processing of each component improves feature extraction and object localization, resulting in greater performance during adverse weather conditions [40]. examines how vehicle exhaust condensation in cold weather impacts LiDAR-based object detectors. It proposes data augmentation and a unique training loss term to address these challenges and achieve more robust 3D object detection.

In [41] they introduced a technique called the Deep Learning Augmented Vision-Based technique (DAVIM) for evaluating material movement under challenging conditions. For accurate and robust sensing of structural model dynamics, DAVIM employs CNN and GAN algorithms. This [42] provides a novel domain adaptive object recognition system that reduces image style and appearance disparities by using both image-level and object-level adaptation for autonomous driving in foggy weather. Additionally, an auxiliary domain and a novel adversarial gradient reversal layer are introduced. An approach was considered in Ref. [43] with the help of combined information from different unfavourable climatic input data to perform a YOLOv8-based object detection using transfer learning. The study discovered that custom datasets facilitated enhanced identification. Also, the usage of additional images with the help of feature-oriented information combined improved object detection performance. An innovative architecture described in Ref. [44] uses Convolutional Neural Networks (CNN) to maintain accurate and excellent-quality topological reconstructions by incorporating ambient data from poor-quality observations. The

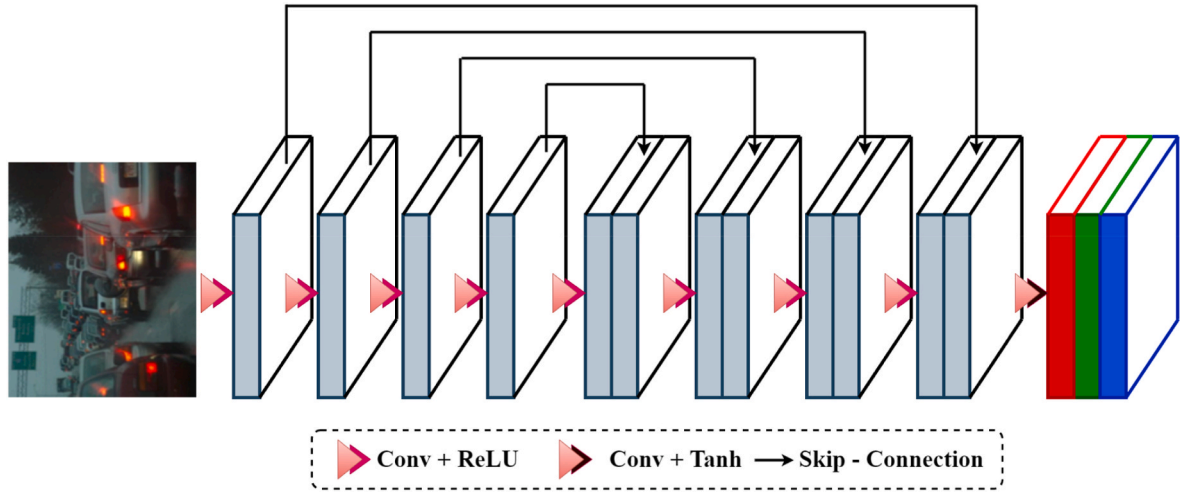


Fig. 3. The pixel-wise depth refinement network.

technique employs information exchange, a non-local attention mechanism, and attention-based multi-scale feature aggregation. This architecture is critical for applications like computational photography, surveillance, self-driving cars, and remote sensing. In a recent study [19], the authors investigated TensorRT to enhance significant outcomes for DL. Their evaluation of TensorRT's performance included assessing interpretation outcome verification, synthesis duration, deduction efficiency, and GPU storage use. Their findings demonstrated that TensorRT can significantly improve inference efficiency metrics while maintaining inference accuracy. This [45] combines the YOLOV4 entity identification approach with more effective identification methodologies. This is 12 % faster than other algorithms and twice as fast as similar algorithms in various applications.

[46] provides an in-depth analysis of automobiles and suggests that researchers should enhance automotive understanding. They need to focus particularly on entity detection under adverse weather conditions, a challenge that may be addressed by information convergence. Datasets were created using different topologies to generate 7 different variants of meteorological circumstances for one dataset and 7 upgraded versions of the same. It helped to distinguish different weather characteristics by analyzing the fluctuations in the datasets. This resulted in improved categorization outcomes in various environmental circumstances [47]. Investigators utilized simulated drops to evaluate the effectiveness of low-light climate enhancement in various racing car circuit scenarios using actual and virtual information. The identification accuracy was tested on a range of meteorological photographs for different durations of the day and various droplet sizes. The main emphasis was on the immediate efficiency of the sensing unit rather than sensory accuracy [48,49]. To boost object identification performance, this [50] explores the integration of noise information into computer vision enhancement models. A noisy data training strategy is developed to improve picture structure improvement by incorporating noise priors into curve estimation training, building upon the Zero-DCE approach. The effect of post-process denoising on object detection is also investigated in this work.

3. Proposed method

3.1. Dataset explanation and annotation

The analysis and detection processes in our proposed system make use of three datasets in total. The datasets used are exclusively dark (Ex-Dark) [51] and contain 7363 images categorized into 12 object classes covering 10 distinct lowlight scenarios, including night scenes, cloudy weather, and indoor areas with limited lighting. From the different

classes, images are selected for the application of autonomous driving object detection and training. Similarly, the second dataset we used was the cityscapes foggy dataset [52], which contains 4000 images from this low-light foggy image that were selected and annotated for the detection process. Finally, we used the DAWN dataset [53] obtained from the Mendeley Data repository. The information provided exhibits a broad range of vehicle movements and varied transportation settings. The database contains 1000 images depicting existing traffic situations categorized into 4 segments according to environmental circumstances. For our work, the Low-light images are specifically selected for interpretation and identification purposes Fig. 2 shows samples of low-light images from the datasets that are used for the training. We implemented a data augmentation technique to artificially expand the training dataset. For dataset annotation, we used Computer vision annotation tools (CVAT), a free and open-source tool with a specific software program developed to label and categorize data for training and enhancing computer vision models. After annotation, export the job dataset to the YOLO1.1 format for training.

3.2. Low light enhancement using pixel-wise depth refinement network

Enhancing low light with a pixel-wise depth refinement is a powerful technique for extracting valuable depth information from images. Focusing on individual pixels and incorporating additional details or residual corrections, significantly improves the accuracy and resolution of depth estimation, leading to a better understanding of the scene's three-dimensional structure for increasing the quality and visibility of images taken under low-light settings. It entails calculating a curve that adjusts the brightness and contrast of each pixel based on its depth in the scene. This enhances the dark regions while preserving the bright regions. There are various ways of estimating the depth curve, including deep neural networks and self-reference schemes. We employ Pixel-Wise Depth Refinement, which views light augmentation as an image-specific curvature prediction job using a deep network and does not rely on paired or unpaired information for training. This is achieved by a sequence of carefully arranged non-orientation loss functions, to train the network without reference pictures. These functions assess the excellence of the augmented image using criteria such as naturalness, noise reduction, and contrast enhancement, encouraging the network to learn efficient curve estimation.

Fig. 3 shows the Pixel-Wise Depth Refinement (PWRN-Net) which collects a reduced-light image as an input. It is a simple CNN composed of nine convolutional layers that will extract and combine characteristics at multiple levels of abstraction and symmetrical concatenation network to learn complementary features from different processing



Fig. 4. Visualization of enhanced image using pixel-wise depth refinement.

streams, leading to a more comprehensive understanding of the underlying patterns. Each layer has a filter size of 64 and 3×3 generative kernels with stride 1, followed by the ReLU activation function. Adding more layers can potentially increase the model's capacity to discover complex associations among input characteristics and depth values, leading to improved accuracy in pixel-wise depth estimation also more convolutional layers can enable the model to extract richer and more diverse features from the low-light image, potentially aiding in better understanding the scene depth. The Tanh activation function follows the last convolutional layer, producing 24 variable mappings across 8 variations, with each variation requiring three curvature variable plots for the 3 bands. Fig. 4 shows the visualization of the images predicted by Pixel-Wise Depth Refinement using Ex-Dark, Foggy Cityscape, and DAWN datasets.

3.3. Lightweight YOLOv8 object detection

3.3.1. Fundamentals of YOLOv8 architecture

Since the first version of the YOLO model was proposed, several studies have updated and iterated on YOLO, and the model's performance has gotten increasingly powerful [54]. The YOLOv8 network design remains separated into three components: backbone, feature enhancement network (neck), and detecting head. YOLOv8 has a similar framework to YOLOv5 but features a few adjustments to the CSP zone, which is now designated as the C2f component. The C2f component consists of a pair of convolutional cross-phase fractional bottlenecks that merge the desired characteristics with situational knowledge to enhance

identification efficiency. YOLOv8 employs an anchor-free framework that uses detached units to independently manage object identification, categorization, and regressive activities. This architecture facilitates the concentration of every component on its specific assignments, leading to a greater overall level of accuracy in the model. The YOLOv8 output layer utilizes the sigmoid functional as the triggering parameter for item ratings. This function accurately represents the probability of an item residing inside the confined area. The SoftMax functional is used to represent category likelihood, which reflects the probability that an item gets assigned within every potential class.

3.3.2. Lightweight YOLOv8 architecture

Fig. 5 shows the architecture of the Lightweight YOLOv8 model. The convolutional block is the most widely used in YOLO. A convolutional block has a 2D convolutional zone, 2D batch regularization, and a SiLU commencement functional. They are all merged into one convolutional block. The next one is the C2f block. This block consists of a convoluted zone, and the resultant characteristic regions will be divided into two separate partitions. One path leads to the obstruction zone, whereas the alternate path leads straight to the concat section. In the C2f block, we can have several bottleneck blocks at the end, followed by another convolutional block. The bottleneck is a set of convolutional zones with a shortcut. Spatial Pyramid Pooling (SPP) is a pooling strategy that allows convolutional neural networks (CNNs) to handle images of varying sizes without losing or distorting the image. The SPPF block in YOLOv8 stands for spatial pyramid pooling fast, which is a variant of SPP with a faster pace. This block begins with a convolutional block at the

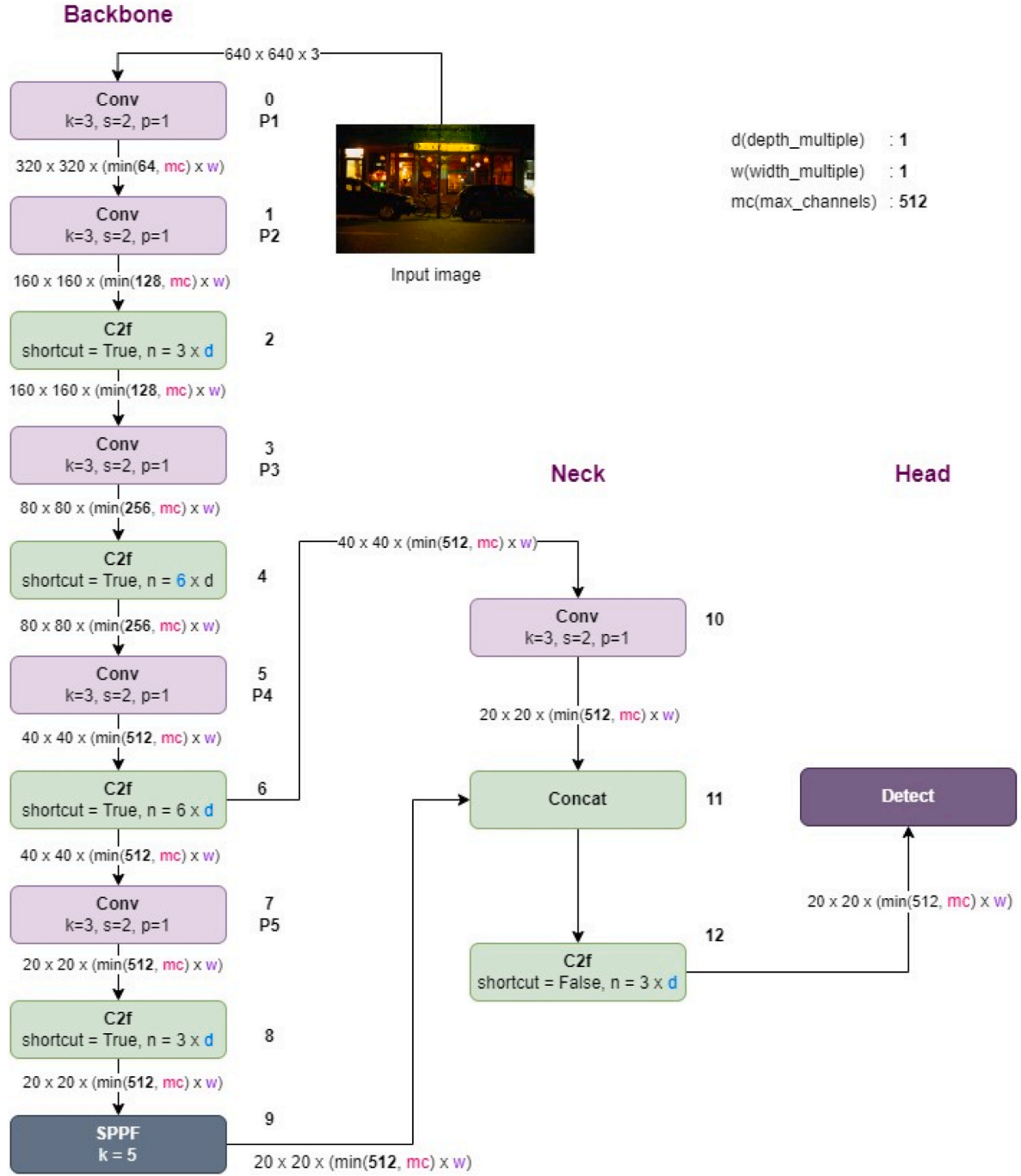


Fig. 5. The Lightweight YOLOv8 architecture.

beginning and is followed by three 2D max pooling layers. The interesting part is that every resulting feature map is concatenated right before the end of SPPF and finally, it ends with a convolutional block as shown in Fig. 6.

The last region is the detect block. This is where the detecting process differs from the previous YOLO version. YOLO is an anchorless model. The predictions occur within the grid cell. The detect block has two tracks. The first track is for bounding box prediction, and the second is for class prediction. Both tracks use the identical block sequence, which includes two convolutional blocks and a single 2D convolutional. The lightweight YOLOv8 architecture is modified by reducing the model size and computational requirements from the YOLOv8 architecture. The parameters depth multiple, width multiple, and max channels are used to control the size and complexity of the model. A 3x3 kernel size (k = 3)

that captures local features while remaining computationally efficient. Using a stride(s = 2) of 2 down samples of the input feature and a padding (p = 1) of one compares that the output feature map has similar geometric dimensions as the input. Short = true indicates residual connections between the input and output of the conv block. shortcut = false indicates no residual connection, while n = 3xd determines channel expansion in the conv block based on a shortcut.

3.4. Improved speed inference using TensorRT

TensorRT is a strong tool for accelerating deep learning inference, providing significant performance benefits over running models directly in frameworks such as TensorFlow or PyTorch. TensorRT is designed primarily for executing neural network models on Nvidia GPU devices,

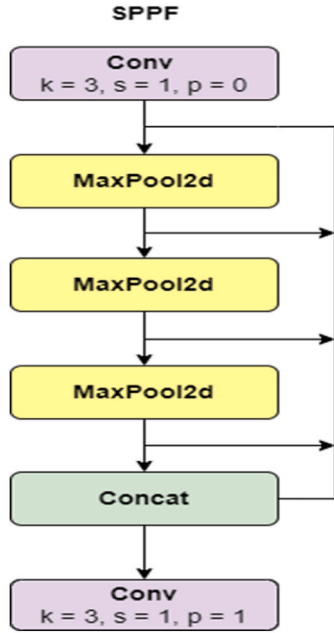


Fig. 6. The spatial pyramid pooling fast block.

to provide high performance while maintaining low latency. TensorRT 8.6.1 is utilized in our proposed system to improve the speed of inference for YOLOv8 object detection on Ex-Dark, Foggy cityscape, and DAWN datasets. The package `tensorrt_lean` eliminates unnecessary or redundant computations in the model inference process and the `tensorrt_dispatch` functions as a lightweight runtime environment for launching pre-built TensorRT engines. The `yolov8` model is converted into an open neural network model (ONNX), and TensorRT does not natively support all of the features and operations used in emerging deep learning frameworks like PyTorch. ONNX serves as an intermediate format for representing the model's computational graph consistently, allowing various frameworks to interpret and process.

3.4.1. Trade-off between speed and accuracy

The balance between velocity and precision gets more intricate proportional to the evolution of the YOLO architecture. YOLOv4 and YOLOv5 simulations comprised innovative elements, including updated networking foundations, enhanced resource replenishment techniques, and enhanced development methodologies. These advancements led to substantial improvements in precision despite severely impacting the real-time functionality of the programs. Starting from YOLOv5, all authorized YOLO variants have optimized the relationship between

rapidity and precision by offering several simulation dimensions that cater to certain workflows and equipment needs. For instance, these iterations often include streamlined models designed for peripheral components, prioritizing quicker execution periods and fewer computing issues above accuracy.

Fig. 7 illustrates a comparison of the various model sizes, from YOLOv5 to YOLOv8. The diagram illustrates the intricacy and efficiency variations across several YOLO framework variants. The initial plot shows the relationship between the range of variables (measured in millions) and the mean average accuracy (mAP) of the COCO verification group. The IOU criteria are from 50 to 95. The data demonstrates a distinct pattern in which augmenting the parameter quantity enhances the framework precision. The second figure illustrates the compromise between the interpretation rapidity (measured as delay on the A100 TensorRT FP16) and the compared system's precision. Every model encompasses distinct levels indicated by n (nano), s (small), m (medium), l (large), and x (extra-large).

4. Experiments and results analysis

4.1. Loss functions

To enhance the process of understanding in-depth curve refinement, we integrate 4 distinct loss processes that are distinguishable and do not rely on each other. This enables us to evaluate the effectiveness of enhanced photos [56].

4.1.1. Spatial consistency loss

Spatial Consistency Loss F_{sp} promotes consistency in the improved image by retaining the dissimilarity among close areas in the input and improved adaptations.

$$F_{sp} = \frac{1}{N} \sum_{i=0}^N \sum_{j \in \mu(i)} (|Y_i - Y_j| - |I_i - I_j|)^2 \quad (1)$$

where N is the number of local zones, and $\mu(i)$ denotes the four regions (right, left, up, and down) centered on region i . The mean intensities of the localized zone in the input and output images are represented as Y and I , respectively.

4.1.2. Exposure control loss

We use exposure control loss to limit under-/overexposure in certain regions. The measurement calculates the difference between the mean magnitude level of a certain area and a predefined degree of good exposure. Here we used $\delta = 0.6$ for the Ex-Dark dataset and $\delta = 0.4$ for foggy and DAWN datasets. It is the exposure control loss function that acts as a reference point for well-exposedness in an image. In general Regions with average intensity lower than 0.6 for the Ex-Dark dataset

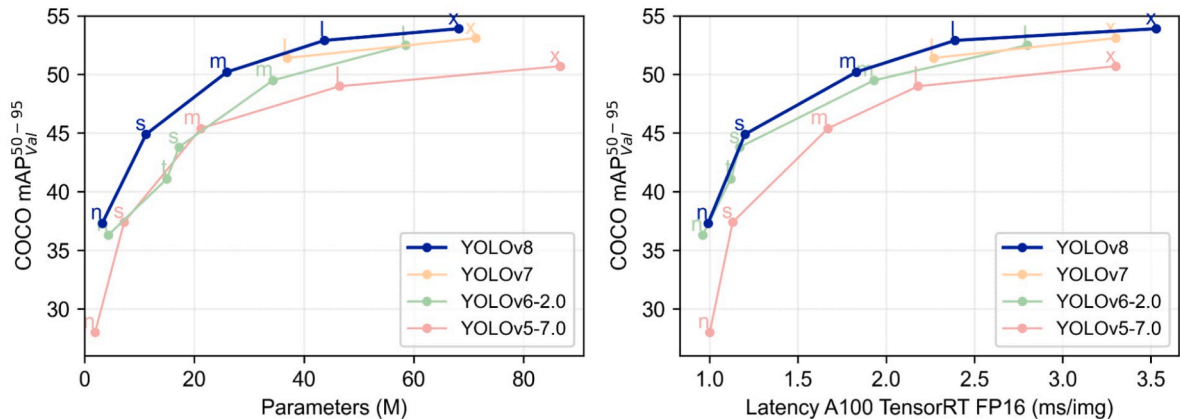


Fig. 7. The Performance comparison of object detection models at various scales, from YOLOv5 to YOLOv8. Plots from Ref. [55].

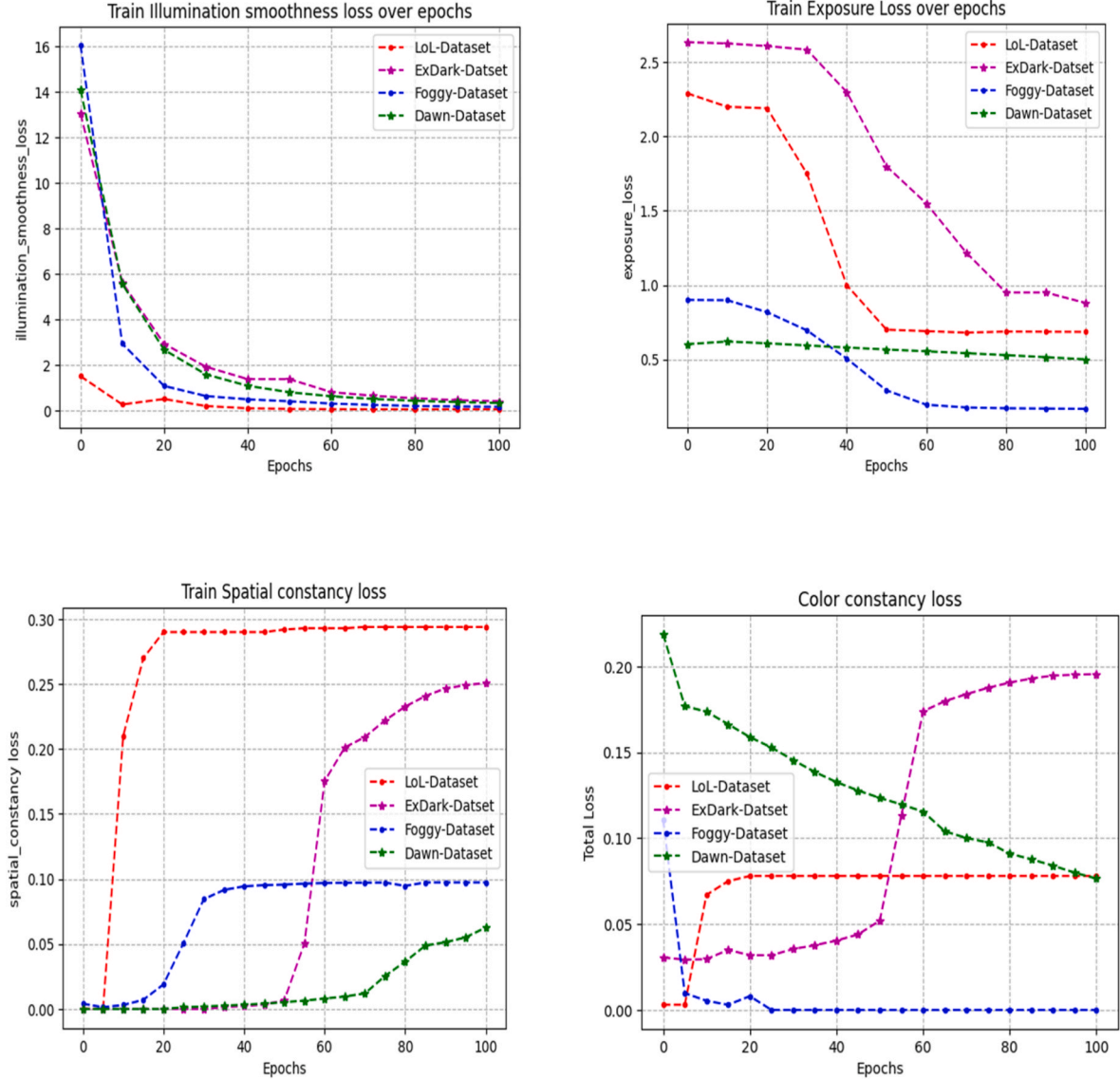


Fig. 8. The graphical representation of non-reference loss functions.

are considered as under-exposed, higher than 0.6 is over-exposed, and exactly equal to 0.6 is perfectly exposed. Similarly, 0.4 for foggy and DAWN datasets. The exposure loss is given in Eq:2 for the Ex-Dark dataset and Eq:3 for Foggy and DAWN dataset

$$F_{exp} = \sum (L(I_{avg}, W) - 0.6)^2 \quad (2)$$

$$F_{exp} = \sum (L(I_{avg}, W) - 0.4)^2 \quad (3)$$

$L(I_{avg}, W)$: Represents a loss function that evaluates the span between the mean intensity value (I_{avg}) of a localized zone and the target well-exposedness value (W).

4.1.3. Color consistency loss (f_{col})

To correct any potential color concerns in the enriched picture and connect the three restructured channels, we use F_{col} can be expressed as follows:

$$F_{col} = \sum_{\forall (p,q) \in \mathcal{E}} (I_p - I_q)^2, \mathcal{E} = \{(R, G), (R, B), (G, B)\} \quad (4)$$

I_p denotes the average channel p intensity value in the output image, with (p,q) representing two distinct conduits.

4.1.4. Illumination smoothness loss

The illumination smoothness loss function attempts to replicate this natural feature during low-light enhancement. It assumes that neighboring pixels on the same surface will have similar illumination intensities. It is given by

$$F_{ill} = \sum ||\nabla I(x)||_2^2 \quad (5)$$

F_{ill} represents the total illumination smoothness loss, $\nabla I(x)$: denotes the gradient of the illumination map I at pixel location x , and $|| \dots ||_2^2$: represents the squared L2 norm.

Fig. 8 shows the graphical representation of F_{sp} , F_{exp} , F_{col} , and F_{ill} . The results have been compared using LoL Dataset [57], Ex-Dark, Foggy cityscape, and Dawn datasets.

4.2. Evaluation metrics

4.2.1. Precision

This refers to the ability to accurately identify true successful samples among all the anticipated positive samples. It is given by:

Table 1

Experimental comparison results in the Exclusive Dark (Ex-Dark) dataset.

Detection Algorithm	Precision	Recall	mAP 0.5	mAP 0.5:0.95
EnlightenGAN + YOLOv5s	0.702	0.601	0.675	0.402
Zero_DCE++ + YOLOv5s	0.692	0.568	0.655	0.383
EnlightenGAN + YOLOv6s	0.661	0.537	0.623	0.336
Zero_DCE++ + YOLOv6s	0.645	0.370	0.598	0.313
EnlightenGAN + YOLOv7	0.702	0.622	0.669	0.387
Zero_DCE++ + YOLOv7	0.681	0.584	0.657	0.354
EnlightenGAN + YOLOv8s	0.709	0.596	0.684	0.465
Zero_DCE++ + YOLOv8s	0.689	0.544	0.654	0.397
EnlightenGAN + DK_YOLOv5	0.747	0.629	0.719	0.467
Zero_DCE++ + DK_YOLOv5	0.723	0.615	0.701	0.456
PWDR + YOLOv8l (ours)	0.812	0.76	0.771	0.532
LPWDR-T YOLOv8l (ours)	0.846	0.812	0.823	0.591

$$P = \frac{TP}{TP + FP} \quad (6)$$

T_p indicates the count of correctly predicted positive samples; F_p reflects the count of incorrectly predicted affirmative items.

4.2.2. Recall

Recall refers to the ability to accurately identify positive samples out of all the projected positive ones. It is given by:

$$R = \frac{TP}{TP + FN} \quad (7)$$

The variable "FN" represents the count of negative samples.

4.2.3. mAP

mAP (mean Average Precision) represents the mean accuracy estimate across all identification classes. It is calculated using the following formula:

$$mAP = \frac{\sum_{n=1}^N AP_n}{N} \quad (8)$$

where N is the total number of classes, and AP_n is the average precision of class n , which corresponds to the region under the Precision-Recall curvature. **mAP@0.5** is the average mean accuracy figure for the IoU variable at an entry value of 0.5. The average of mAP is denoted as **mAP@0.5:0.95**, which represents the range of IoU variable dimensions [0.5:0.05:0.95].

4.3. Comparison of overall experimental results

4.3.1. Training configuration and hyperparameter tuning

To evaluate the generalizability of our Lightweight YOLOv8-based object detection approach, we conducted experiments on three distinct datasets Ex-Dark, Foggy cityscape, and DAWN dataset. To ensure a consistent evaluation across these datasets, we employed a common training configuration for Lightweight YOLOv8. This configuration included:

Device: We utilized a single GPU for training and CUDA Version:

12.2. Epochs: The model was trained for a maximum of 100 epochs, with early stopping if the validation performance did not improve for 30 consecutive epochs (patience). Workers: We employed 2 worker processes to handle data loading and preprocessing tasks. Batch Size: A batch size of 12 images was used for training. Learning Rate: The initial learning rate (lr0) was set to 0.01, and a learning rate scheduler was employed to gradually reduce it to a final learning rate (lrf) of 0.01 during training. Momentum and Weight Decay: Momentum (0.95) and weight decay (0.0001) were used to improve model convergence and generalization. Warmup Strategy: To improve convergence during the initial training phase, a warmup strategy was implemented with 10 warmup epochs, a warmup momentum of 0.5, and a warmup bias learning rate of 0.1. Optimizer: The Stochastic Gradient Descent (SGD) optimizer was chosen for model training.

4.3.3. Performance comparison for Foggy Cityscape dataset

Table 2 evaluates the working of different entity identification approaches on the Foggy Cityscapes dataset, encompassing diverse object categories like Person, Bicycle, Motorcycle, Rider, Bus, Car, Train, and Truck. Additionally, it provides an overall mAP metric combining all categories. LPWDR-T YOLOv8l achieved the highest overall performance with an mAP of 52.1 %, outperforming previous methods across most categories. SADA and MIC-SADA based on YOLOv3 achieved notable results. These methods were close contenders, particularly in categories like Car and Rider detection. SIGMA excelled in Car detection but struggled with some other categories. While achieving the highest mAP for Cars (60.3 %), its performance wasn't consistent across all objects.

4.3.2. Performance comparison for Ex-Dark dataset

Table 1 presents a comparison of the effectiveness of several item identification approaches on a job involving brightening low-light imagery. The algorithms combine image enhancement techniques (EnlightenGAN, Zero_DCE++, PWDR) with different object detection models (YOLOv5s, YOLOv6s, YOLOv7, YOLOv8s, DK_YOLOv5, PWDR + YOLOv8l LPWDR-T YOLOv8l). Performance is measured using Precision, Recall, mAP 0.5 (IoU threshold of 0.5), and mAP 0.5:0.95 (IoU thresholds between 0.5 and 0.95). LPWDR-T YOLOv8l (Lightweight object detection for Pixel-Wise Depth Refinement and TensorRT optimization) achieved the highest overall performance with an accuracy of 0.846, Recall of 0.812, mAP 0.5 of 0.771, and mAP 0.5:0.95 of 0.591. This method combines a Pixel-Wise Depth Refinement (PWDR) with Lightweight YOLOv8l for image enhancement and object detection, respectively. EnlightenGAN generally outperformed Zero_DCE++ across different models. EnlightenGAN consistently had higher Precision and mAP values, indicating better object localization and accuracy. Performance varied across YOLO versions. While EnlightenGAN + YOLOv7 [64] achieved the highest mAP 0.5 (0.669), EnlightenGAN + YOLOv8s had the highest mAP 0.5:0.95 (0.465), suggesting better performance for detecting small objects. EnlightenGAN + DK_YOLOv5 [63] showed promising results, achieving the second highest Precision (0.747) and mAP 0.5:0.95 (0.467) among all combinations. This suggests potential for further exploration of this method.

Table 2

The Experimental comparison results in the Foggy Cityscapes dataset.

Method	Person	Bicycle	Mticycle	Rider	Bus	Car	Train	Truck	mAP
DA Faster [51]	38.3	40.4	19.7	28.5	29.2	43.4	23.7	32.7	32.0
SW-DA [58]	43.8	44.3	21.0	28.0	31.8	48.9	28.9	35.8	35.3
SC-DA [59]	42.5	42.1	26.8	26.5	33.8	52.1	29.2	34.5	35.9
MTOR [60]	30.6	35.6	28.3	41.4	38.6	44.0	40.6	21.9	35.1
SIGMA [61]	44.0	40.6	31.7	43.9	50.4	60.3	51.5	31.6	44.2
SADA [62]	48.5	45.4	32.4	52.6	50.3	62.1	31.5	29.5	44.0
MIC -SADA [63]	50.9	47.5	40.6	55.3	52.4	67.0	33.7	33.9	47.6
PWDR + YOLOv8l (ours)	50.7	48.6	42.7	57.4	55.8	70.2	48.6	39.8	51.72
LPWDR-T YOLOv8l (ours)	51.1	48.2	43.2	57.4	57.0	71.0	48.7	40.2	52.1

Table 3

Comparison outcomes in the DAWN dataset.

Model	Person mAP	Mt cycle mAP	Car mAP	mAP
YOLOv5	87 %	73 %	86 %	82 %
YOLOv7	89 %	79 %	90 %	86 %
PWDR + YOLOv8l (ours)	88 %	62 %	92 %	90 %
LPWDR-T YOLOv8l (ours)	90 %	68 %	96 %	93 %

Table 4

The Inference speed using TensorRT.

Model	Ex-Dark Dataset	Foggy cityscape Dataset	DAWN Dataset
Inference Speed			
YOLOv8l	106.4 ms	96.6 ms	107 ms
PWDR + YOLOv8l (ours)	97.5 ms	72.6 ms.	99.5 ms
LPWDR-T YOLOv8l (ours)	25.0 ms	24.4 ms	24.9 ms

4.3.4. Performance comparison for the DAWN dataset

Table 3 evaluates the effectiveness of the 4 entity identification approaches on the DAWN dataset, evaluating them across three object categories: Person, Motorcycle, and Car. Finally, it provides an overall mAP metric combining all categories. LPWDR-T YOLOv8l achieved the highest overall performance with a Person mAP of 90 %, Bicycle mAP of 68 %, Car mAP of 96 %, and overall mAP of 93 %. This method combines

Pixel-Wise Depth Refinement (PWDR) with Lightweight YOLOv8l for image enhancement and object detection, respectively. YOLOv8l models generally outperformed other methods. Both PWDR + YOLOv8l and LPWDR-T YOLOv8l variants using YOLOv8l architecture achieved higher overall mAP compared to YOLOv5 and YOLOv7.

4.3.5. Inference speed comparison for all datasets

Table 4 summarizes the inference speed improvements achieved by utilizing TensorRT 8.6.1 for YOLOv8 object detection on three challenging datasets: Ex-Dark, Foggy Cityscape, and DAWN. Implementing TensorRT with LPWDR YOLOv8l resulted in substantial performance gains across all datasets. Compared to the base YOLOv8l model, inference speed improved by:76.4 % on Ex-Dark (from 106.4 ms to 25.0 ms),72.9 % on Foggy Cityscape (from 96.6 ms to 24.4 ms),72.7 % on DAWN (from 107 ms to 24.9 ms) The proposed LPWDR YOLOv8l model consistently showed a greater speed up with TensorRT compared to the baseline YOLOv8l, highlighting the potential of optimizations for complex models. The speedup achieved with TensorRT was remarkably consistent across all three datasets, indicating its effectiveness for various challenging conditions.

Fig. 9. (a) Shows the lowlight input images, (b) Object detection in low-light using YOLOv8 Model and (c) explores the potential of combining pixel-wise depth refinement and lightweight object detection for improved performance in challenging low-light conditions. We utilize three datasets: Ex-Dark, Foggy Cityscape, and DAWN, each containing low-light images. For each dataset, we present a visual comparison using a 3x3 grid. In Fig. 9(a), the original low-light image



Fig. 9. (a) Low-light Input images from the datasets, (b) Object detection in low-light using the YOLOv8 Model, and (c) Object detection using the LPWDR-T method.

showcased the challenging visual environment. Fig. 9(b) displays the outcomes of entity identification using YOLOv8 model on the low-light image, showcasing the model's proficiency in recognizing items. Lastly, Fig. 9(c) shows the improved image produced by our suggested approach of refining depth on a pixel-by-pixel basis, paired with the Lightweight efficient YOLOv8 algorithm (LPWDR-T) for detecting objects. Our findings reveal that pixel-wise depth refinement leads to improved object detection accuracy in low-light conditions across all three datasets (Ex-Dark, Foggy Cityscape, DAWN). This observation validates the efficiency of our suggested method for enhancing entity identification performance in challenging visual environments. However, there are some potential limitations to consider: Depth estimation, especially in low-light conditions, can be inherently challenging. Factors like shadows or lack of texture can lead to inaccurate depth predictions, potentially impacting object detection performance. Also, the TensorRT optimization improves inference speed, but real-time performance might still be limited on devices with very low computational resources. In future work we will continue Implementing attention mechanisms within the object detection model to dynamically focus on regions with high-quality depth information, leading to better object recognition.

5. Conclusion

The research outlined in this manuscript showcased an innovative method for enhancing object identification in low-light conditions. We combined three key elements: lightweight object detection using a YOLOv8 model for efficiency on resource-constrained devices, pixel-wise depth refinement with modified depth curve estimation for detailed scene depth information, particularly in low-light scenarios, and TensorRT optimization for faster inference and real-time performance. Our proposed approach demonstrates the potential of this combination to achieve improved object detection accuracy, with our experiments revealing higher detection accuracy, precision, and recall rates compared to existing methods. Additionally, we observed a 72 % improvement in inference speed compared to previous approaches, further demonstrating the efficiency of our methodology. Our future work will focus on addressing these limitations and enhancing the system's robustness. We will explore incorporating attention mechanisms within the object detection model. By dynamically focusing on regions with high-quality depth information, we aim to improve object recognition even in areas where depth estimation might be less accurate. Additionally, we can investigate techniques like model quantization to further reduce the model size and enable deployment on devices with even lower computational resources. By addressing these limitations and exploring these future directions, we can refine our approach for a more robust and generalizable low-light object detection system capable of real-time performance on a wider range of devices.

CRedit authorship contribution statement

K. Vinoth: Writing – original draft. **Sasikumar P:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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